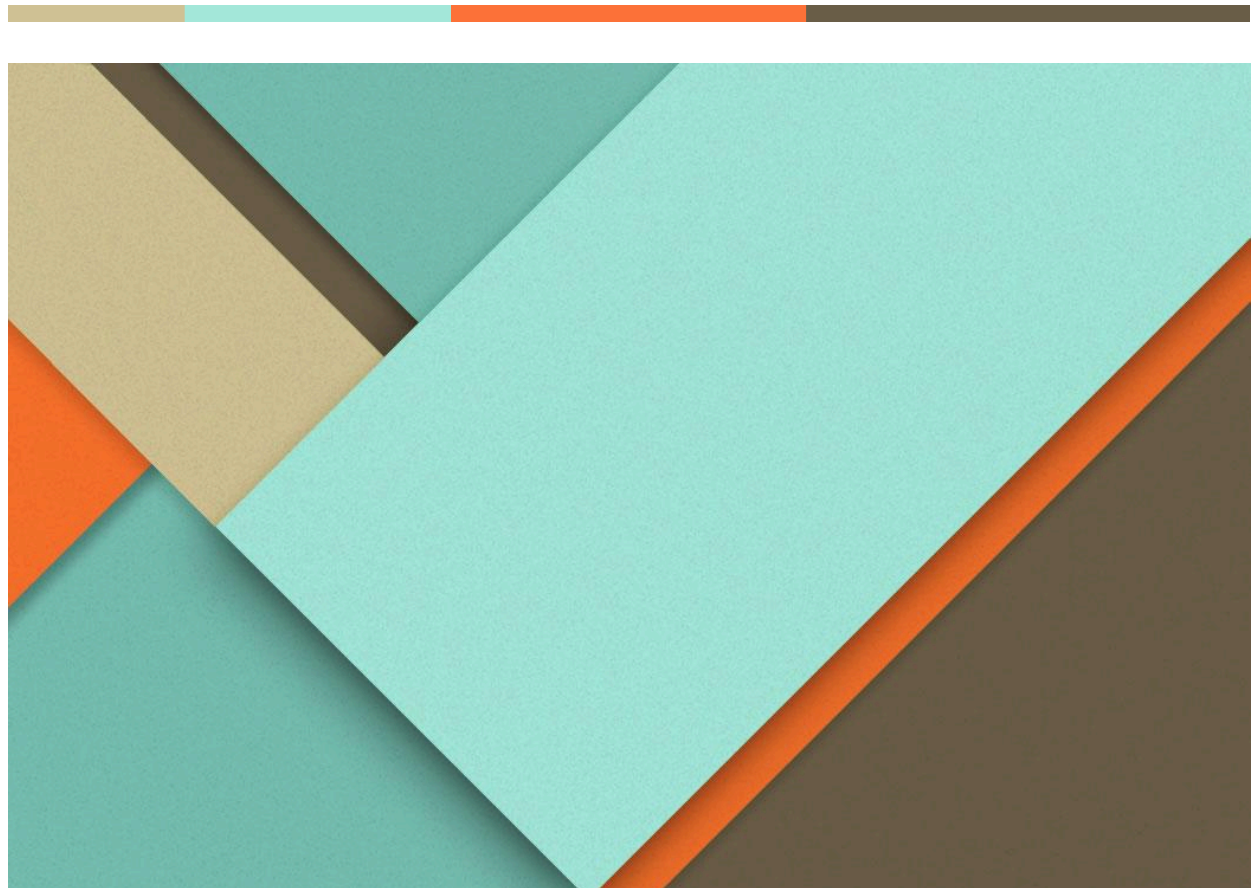




SmartExam

09.04.2026

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-  GitHub: <https://github.com/singhrajvardhan>
-  Portfolio: www.rajvardhansingh.in
-  Live Project: <https://examsystem.liveblog365.com>

Overview

SmartExam is a production-ready, AI-powered online examination platform engineered to deliver secure, scalable, and integrity-driven assessments in remote environments. It combines intelligent proctoring with a robust exam engine to replicate the discipline of in-person examinations in a fully digital workflow. Designed with a modular architecture, the system cleanly separates frontend, backend, AI processing, and database layers, enabling maintainability, extensibility, and enterprise-grade scalability across institutions and certification platforms.

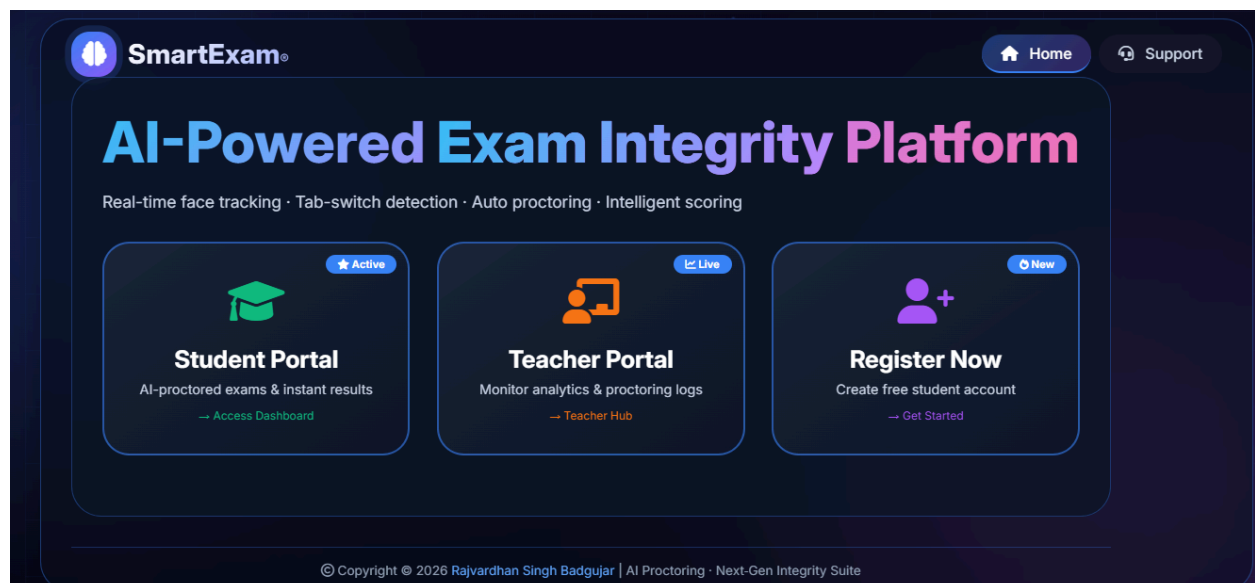
At its core, SmartExam integrates advanced **AI-based proctoring**, leveraging technologies like OpenCV and TensorFlow for real-time face detection, liveness verification, and behavioral analysis. The platform actively detects anomalies such as tab switching, screen absence, and suspicious interactions, generating instant alerts and logs. The exam system supports automated evaluation, negative marking, and dynamic question management, while secure authentication is handled through token-based mechanisms to ensure data protection and session integrity.

Built using a modern full-stack approach with JavaScript, Python, PHP, and Java, SmartExam utilizes PostgreSQL for reliable data management and Socket-based communication for real-time monitoring. The project follows strict Git workflows with enforced pull requests, code reviews, and branch protection to maintain high code quality. Developed collaboratively by a 20-member team across Cyber Security and Computer Science domains, SmartExam reflects a strong emphasis on system design, security, and real-world deployment, positioning it as a comprehensive solution for next-generation digital assessments.

It supports multiple user roles including:

- [Student](#)
- [Teacher \(Examiner\)](#)
- [Admin \(System Controller\)](#)

The platform ensures seamless exam conduction with **automated workflows, analytics, and reporting systems**.



Key Features

Security & Proctoring

SmartExam ensures a secure examination environment through multiple layers of protection. It detects tab switching and restricts copy-paste actions to minimize unfair practices. The system uses AI-based auto-proctoring to monitor candidate behavior, along with optional camera-based activity tracking. Any suspicious activity is recorded instantly, allowing administrators to review violations in real time and take appropriate action.

Smart Exam Engine

The platform offers a flexible and efficient exam engine designed for accuracy and fairness. Questions can be randomized for each student to prevent answer sharing. It supports automatic evaluation for objective-type questions, including negative marking where required. Results are processed instantly, reducing manual effort and enabling faster decision-making for educators.

Analytics & Reporting

SmartExam provides clear and actionable insights into student performance. It generates detailed reports including subject-wise analysis, accuracy tracking, and rank comparisons. These analytics help educators identify strengths, weaknesses, and overall trends, making it easier to improve teaching strategies and student outcomes.

Communication System

The system includes a built-in communication module to keep students and administrators informed. It automatically sends important emails such as registration confirmations, roll numbers, and application IDs. After exam completion, results and performance details are distributed via email, ensuring a smooth and professional communication process.

 **SmartExam®**

[Home](#) [Student](#) [Teacher](#) [Support](#)

24/7 Support Center

Our AI-powered support team is here to help you. Submit a ticket and we'll get back to you within 2 hours.

Submit a Support Ticket

 Full Name *

 Email Address *

 Ticket Category

 Technical Issue



 Message *

 Submit Ticket

 Your data is encrypted and secure · We reply within 2h

Help & Resources

**Response Time**

Average response: < 2 hours during business hours.
Emergency support available 24/7.

**AI-Powered Assistance**

Our intelligent system categorizes and prioritizes tickets for fastest resolution.

**Knowledge Base**

Check our FAQ and documentation for instant answers to common questions.

**Alternative Contact**

**Email Support**

rajvardhanocoder@gmail.com

**Live Chat**

Available 9 AM - 9 PM IST



System Architecture

SmartExam is structured around a **multi-portal architecture** that separates responsibilities, improves security, and ensures smooth system operations. Each portal is designed for a specific user role, allowing better control, scalability, and user experience across the platform.



Core Portals



Public Gateway

This is the entry point of the system where users can access general information about exams, register, and navigate to login pages. It acts as the first interaction layer, ensuring a clean and user-friendly onboarding experience for students and institutions.



Authentication Hub

The Authentication Hub manages secure login, registration, and identity verification. It handles user credentials, session management, and token-based authentication, ensuring that only authorized users can access their respective dashboards and exam environments.



Student Command Center

This portal is designed for students to manage their entire exam journey. It allows them to view upcoming exams, attend tests, track their performance, and receive real-time alerts during exams. The interface is optimized for simplicity while maintaining strict exam controls.



Educator & Proctor Suite

This module enables teachers and proctors to create exams, manage questions, monitor live exam sessions, and review student behavior. It provides tools for real-time supervision, violation tracking, and performance evaluation, ensuring effective exam management.



Administration Core

The Administration Core acts as the central control system of SmartExam. It manages users, system configurations, security policies, and overall platform operations. Admins can monitor system health, control permissions, and ensure smooth coordination between all modules.



Navigation Atlas (System Routes)



Public Gateway

The Public Gateway defines the external navigation layer of SmartExam, offering a structured and informative interface for all users before authentication. It ensures clarity, accessibility, and trust by presenting essential information about the platform in a well-organized manner.

- [Home](#) – Home Preview
Serves as the landing page, providing a high-level overview of the platform, key features, and a preview of the system interface to help users quickly understand its capabilities.
- [About System](#)
Explains the purpose, vision, and core functionality of SmartExam, giving users insight into how the platform ensures secure and efficient online examinations.
- [FAQ & Help Center](#)
Offers answers to common queries and guides users through typical processes, reducing support dependency and improving user experience.
Provides direct communication channels for users to reach out for assistance, report issues, or seek clarification, ensuring responsive support.
- [Privacy Policy](#)
Clearly outlines how user data is collected, stored, and protected, reinforcing transparency and compliance with data protection standards.
- [Terms of Service](#)
Defines the rules, responsibilities, and usage guidelines for all users, establishing a clear legal framework for platform usage.



Authentication Hub

The Authentication Hub serves as the secure access layer of SmartExam, ensuring that only verified users—students, teachers, and administrators—can access the system. It delivers a seamless and role-based login experience while maintaining strong security through credential validation, session management, and access control.

The Student Login module enables quick entry into the exam environment, while the Teacher and Admin Login modules provide access to exam creation, monitoring, evaluation, and system management features. The Registration process captures essential user details and generates unique identifiers such as Application ID and Roll Number for students, ensuring proper onboarding and identity tracking.



Student Command Center

The Student Command Center acts as a centralized workspace where students can manage their entire examination lifecycle in a structured and user-friendly interface. It is designed to balance simplicity with strict exam controls, ensuring both ease of use and system integrity.



Student Registration Page

The Registration Page is the first step for new users to join the SmartExam platform. It is designed with simplicity and validation-driven inputs to ensure accurate data collection and secure account creation.

♦ Key Functionalities:

- **Personal Information Capture**
Collects essential details such as Full Name, Email Address, Contact Number, and Date of Birth etc for identity creation.
- **Unique ID Generation**
Automatically generates Application ID and Roll Number, ensuring each student is uniquely identifiable in the system.
- **Email Verification System**
Sends a confirmation email to validate the user's email address and activate the account.
- **Error Handling & Validation**
Real-time form validation prevents incorrect or incomplete submissions, improving data accuracy.
- **User-Friendly Interface**
Clean layout with guided input fields, making registration quick and accessible even for first-time users.

Student Registration

Full Name *

Father's Name *

Mother's Name *

Email *

Your credentials will be sent to this email

Phone Number *

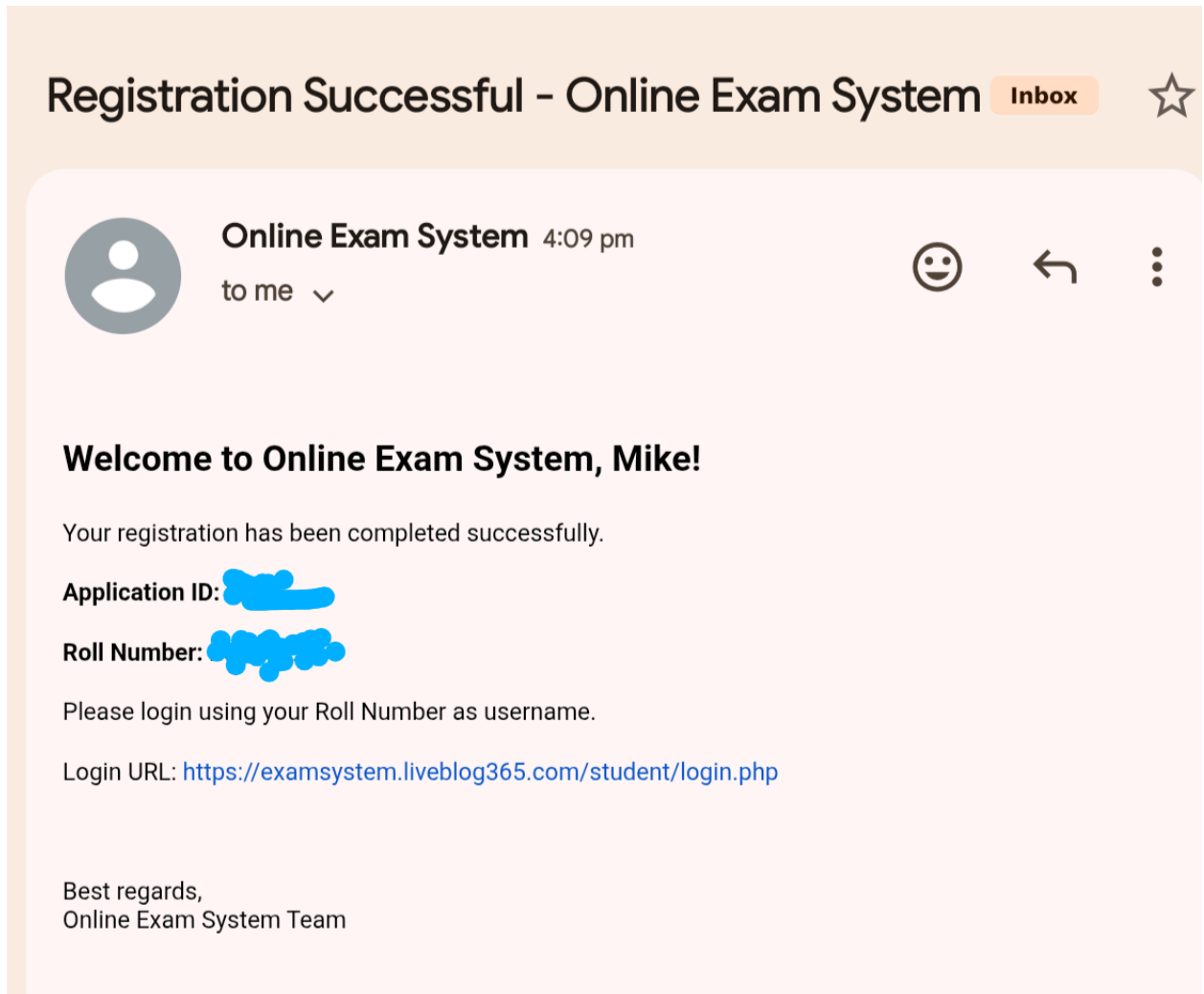
Address *

Date of Birth *

dd-mm-yyyy

Register

Already have an account? [Login here](#)



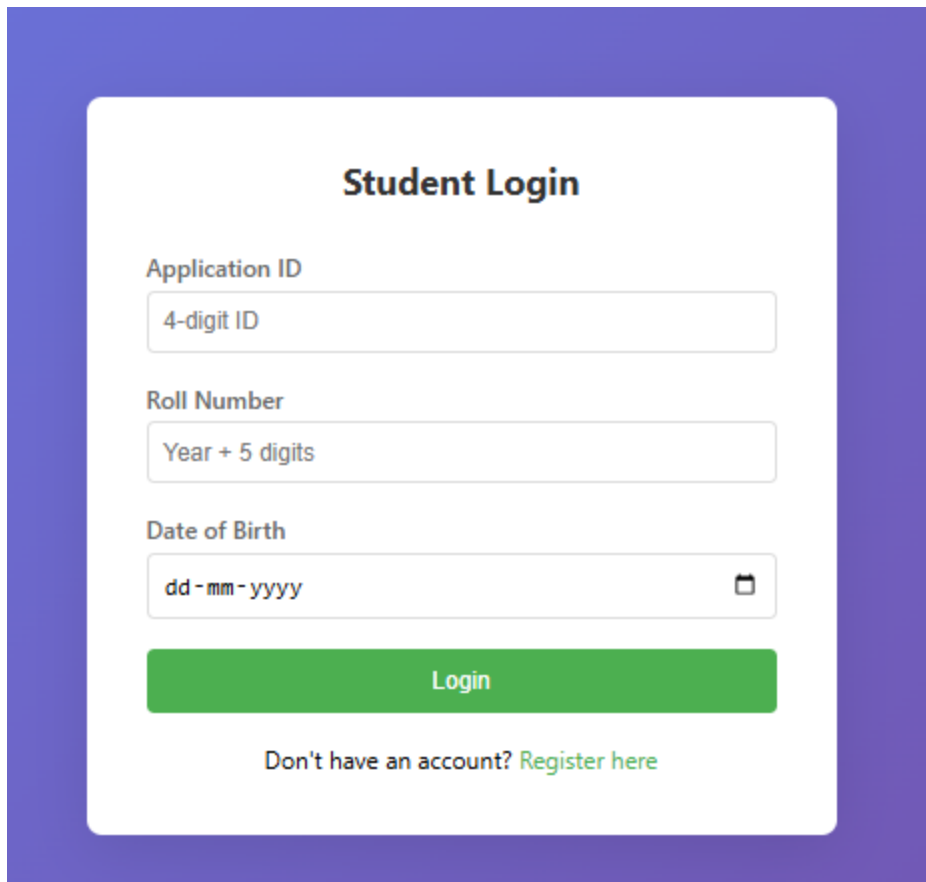
Student Login Page

The Login Page provides secure and efficient access to the SmartExam system for registered users. It is optimized for both usability and security.

♦ Key Functionalities:

- **Credential-Based Login**
Students can log in using Roll Number ,Application ID and Date of Birth for flexible access.
- **Session Management**
Maintains secure user sessions and prevents unauthorized access.

- **Multi-Layer Security Checks**
Includes protection against brute-force attempts and unauthorized login activities.
- **Quick Access Design**
Minimal and fast-loading interface to allow students to enter the exam environment without delay.
- **Error Feedback System**
Provides clear messages for invalid credentials or login issues, improving user experience.

A screenshot of a 'Student Login' form. The form is white with rounded corners and is centered on a purple background. It contains three input fields: 'Application ID' with a placeholder '4-digit ID', 'Roll Number' with a placeholder 'Year + 5 digits', and 'Date of Birth' with a placeholder 'dd - mm - yyyy' and a calendar icon. Below the fields is a green 'Login' button. At the bottom, there is a link that says 'Don't have an account? Register here'.

[Dashboard Overview](#)

Upon login, students are welcomed with a **personalized dashboard** displaying real-time system information such as date, time, and session status. A greeting message (e.g., *"Welcome back, Rajvardhan!"*) enhances user engagement, while system indicators like **performance status (Stable)** provide quick insights into exam readiness.

The dashboard prominently displays **student identity details**, including:

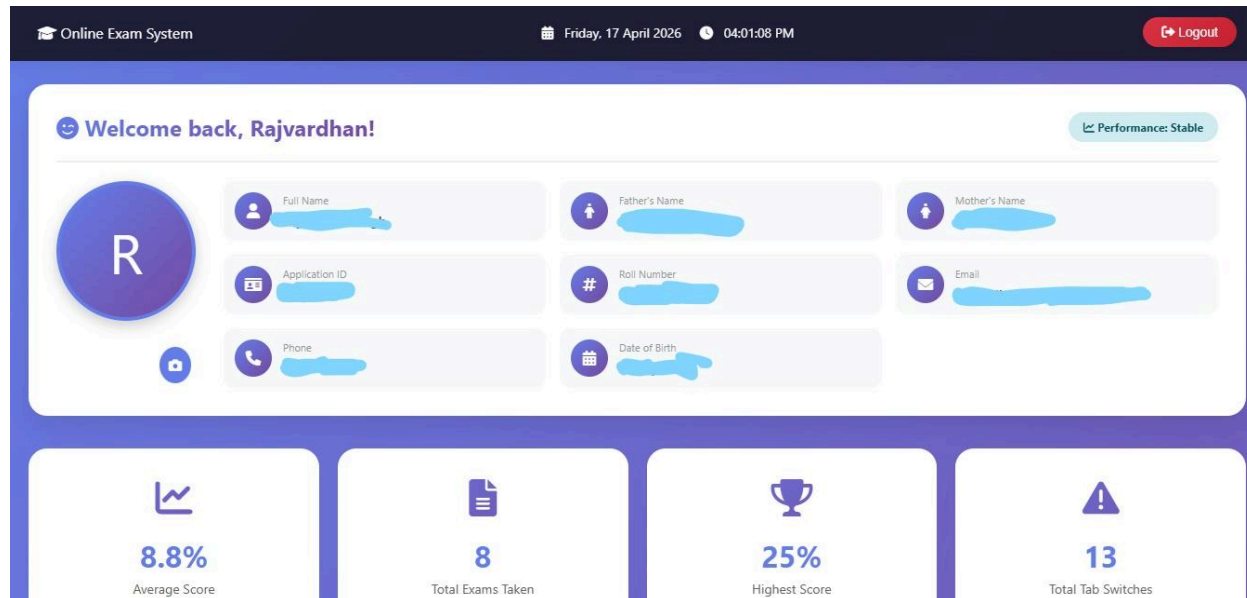
- Full Name, Father's Name, Mother's Name
- Application ID & Roll Number (system-generated unique identifiers)
- Email, Phone Number, and Date of Birth

This ensures transparency and allows students to verify their registered information instantly.

In addition, key performance metrics are highlighted in a **compact analytics section**, including:

- Average Score (e.g., 8.8%)
- Total Exams Taken (e.g., 8)
- Highest Score (e.g., 25%)
- Total Tab Switches (e.g., 13)

These metrics provide a quick snapshot of overall academic performance and exam behavior.



Profile Management

The **Profile Management module** allows students to securely manage and review their personal information. Users can view essential details such as name, contact information, and academic identifiers, ensuring data accuracy.

It also enables updates to certain fields and supports **security features like credential management**, ensuring that account information remains protected and up to date. This module plays a critical role in maintaining user identity integrity across the system.

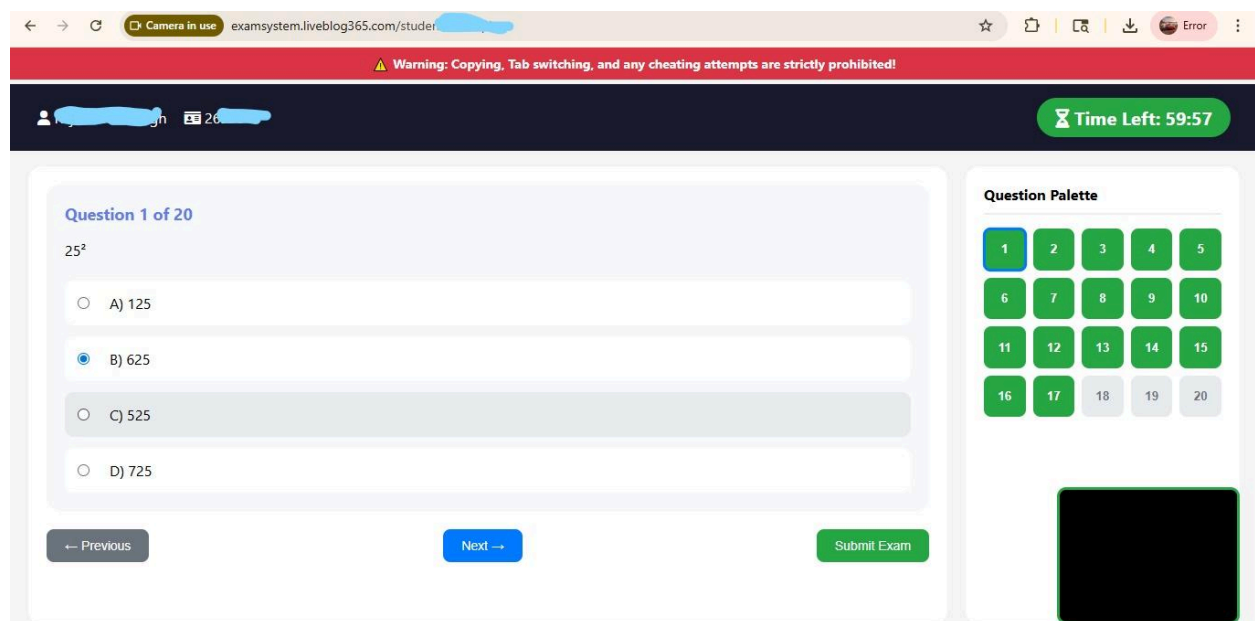
Examination Zone

The **Examination Zone** is the core functional area where students participate in exams under a controlled environment. It displays key exam details such as:

- Total Questions (e.g., 20 Questions)
- Duration (e.g., 40 Minutes)
- Negative Marking rules

Students can initiate exams through a **“Start Examination”** action, with clear system instructions such as ensuring a stable internet connection and active camera (if required).

The exam interface is designed to be **timer-driven and auto-submitting**, ensuring fairness and consistency. Built-in monitoring features track behavior during the exam, reinforcing system integrity.



Results & Analytics

This section provides **detailed performance tracking and historical insights**. After each exam, students receive an **instant result summary**, including:

- Completion date and time

- Score percentage (e.g., 25%)
- Correct answers (e.g., 5/20)
- Tab switching activity (e.g., 4 times)

The **Recent Exam History table** offers a structured view of past attempts, including:

- Date & time of exam
- Score and accuracy
- Tab switches recorded
- Rank achieved

These insights help students evaluate their progress, identify patterns, and improve future performance.

Last Exam Result				
Completed:	Score:	Correct Answers:	Tab Switches:	
13 Apr 2026, 09:21 AM	25%	5 / 20	4	

Recent Exam History				
Date	Score	Correct/Total	Tab Switches	Rank
13 Apr 2026, 09:21 AM	25%	5 / 20	4	#1
13 Apr 2026, 09:20 AM	25%	5 / 20	3	#1
13 Apr 2026, 07:48 AM	0%	0 / 20	3	#6
13 Apr 2026, 07:48 AM	0%	0 / 20	3	#6
13 Apr 2026, 01:08 AM	10%	2 / 20	0	#3

Certificates

The **Certificates module** allows students to download **digitally generated certificates** upon successful exam completion. These certificates serve as official proof of participation and performance, making them useful for academic records, job applications, and professional validation.

The system ensures that certificates are **secure, verifiable, and instantly accessible**, enhancing the overall value of the platform.

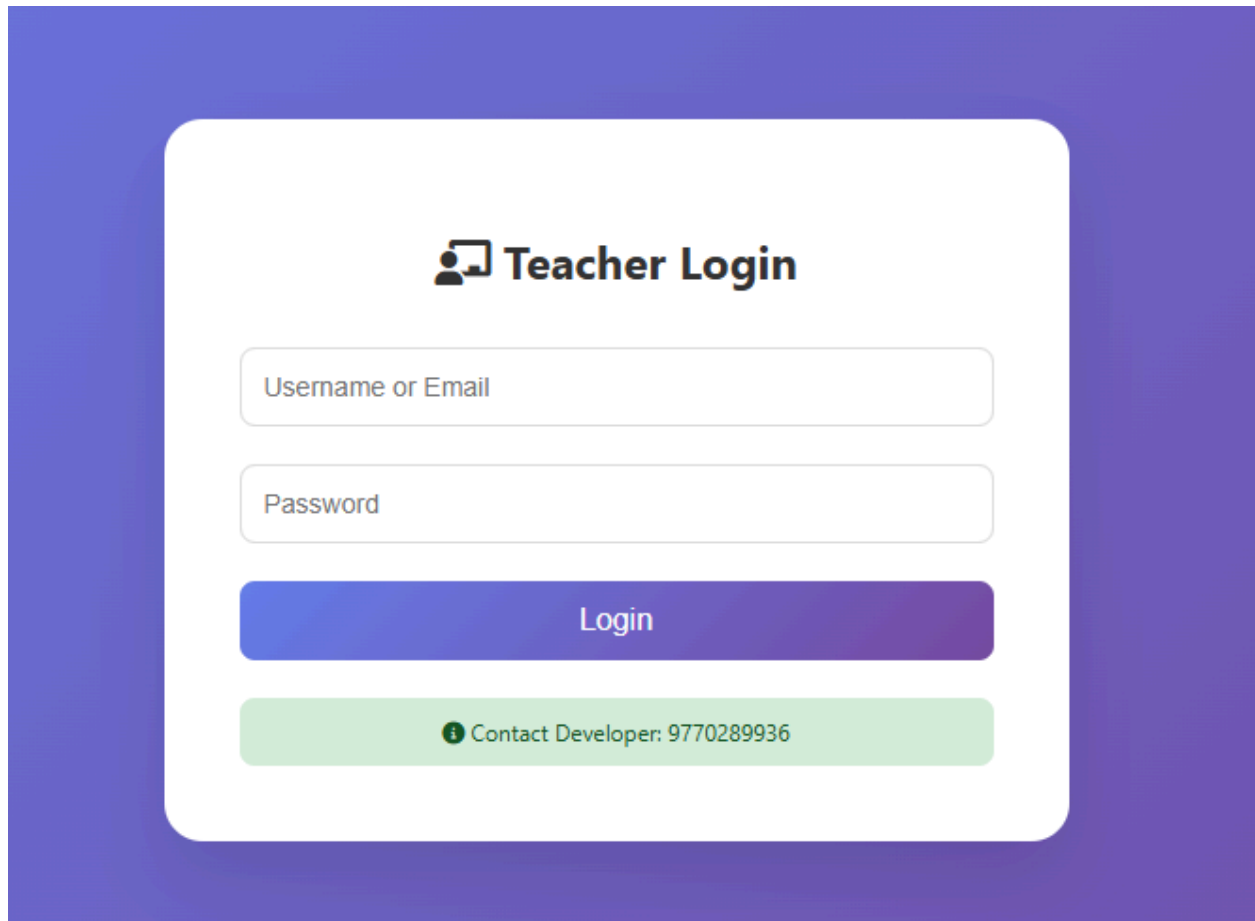
Teacher Flow

The Teacher Flow in SmartExam is designed to provide educators with a structured, secure, and efficient workflow for managing the complete examination lifecycle. It empowers

teachers to control exams, monitor student activity, and evaluate performance from a single unified interface.

Secure Login Access

Teachers access the system using **admin-provided credentials**, ensuring that only authorized personnel can manage academic operations. The authentication process is secure and role-based, granting appropriate permissions based on responsibilities.



The image shows a 'Teacher Login' interface. It features a white rounded rectangle centered on a purple gradient background. At the top of the white box is a black icon of a person at a computer, followed by the text 'Teacher Login'. Below this are two input fields: 'Username or Email' and 'Password'. A blue-to-purple gradient 'Login' button is positioned below the password field. At the bottom of the white box is a light green button with a green information icon and the text 'Contact Developer: 9770289936'.

Application ID	Roll Number	Full Name	Email	Phone	Registered On	Actions
[Redacted]	[Redacted]	Mike	[Redacted]@gmail.com	[Redacted]	17 Apr 2026	Edit Delete Results
[Redacted]	38	System Administrator	[Redacted]@gmail.com	[Redacted]	14 Apr 2026	Edit Delete Results

Exam & Resource Management

Teachers are equipped with powerful tools to manage all critical components of the examination process:

- Student Management**
 Add, view, and manage student records, ensuring accurate mapping between candidates and exams.
- Question Management**
 Create, edit, and organize question banks with support for structured exam patterns and randomization.
- Exam Configuration**
 Define exam settings such as duration, marking scheme (including negative

marking), and rules to ensure standardized assessments.

Teacher Dashboard | Welcome, System Administrator [Logout](#)

Students Exam Results **Questions** Security Logs

[+ Add New Question](#)

ID	Question	Option A	Option B	Option C	Option D	Correct	Marks	Actions
1	25♦...	125	625	525	725	B	4	Edit Delete
2	?64 =?...	6	7	8	9	C	4	Edit Delete
3	Who was the first Prime Minister of India?...	Mahatma Gandhi	Jawaharlal Nehru	Sardar Patel	Subhas Chandra Bose	B	4	Edit Delete
4	In which year did India become a republic?...	1947	1948	1949	1950	D	4	Edit Delete
5	Who built the Red Fort?...	Akbar	Babur	Shah Jahan	Aurangzeb	C	4	Edit Delete

Real-Time Monitoring

During live exams, teachers can actively supervise sessions through a dedicated monitoring interface:

- **Live Exam Sessions**
Track student activity in real time, ensuring smooth exam execution.
- **Security Logs & Alerts**
View detailed logs of suspicious activities such as tab switching or unusual behavior, enabling quick intervention if needed.

Teacher Dashboard | Welcome, System Administrator [Logout](#)

Students Exam Results Questions **Security Logs**

43
Total Tab Switches

5
Students with Camera Issues

Tab Switch Logs

Student ID	Total Switches	Actions
	19 High Risk	View Details
	13 High Risk	View Details

Performance Analysis

Post-exam, teachers gain access to detailed analytics that support data-driven evaluation:

- Review individual and group performance
- Identify trends, strengths, and weak areas
- Compare student rankings and accuracy levels

This helps educators improve both assessment quality and teaching strategies.

Result Communication

The platform simplifies communication by allowing teachers to **send result reports directly to students via email**. These reports include scores, performance insights, and relevant exam details, ensuring timely and professional result distribution.



The screenshot shows a 'Teacher Dashboard' with a dark header bar containing a user icon, the text 'Teacher Dashboard | Welcome, System Administrator', and a red 'Logout' button. Below the header is a navigation bar with tabs: 'Students', 'Exam Results' (active), 'Questions', and 'Security Logs'. The main content area displays a table of exam results for two students.

Student Name	Roll Number	Score	Correct/Wrong	Tab Switches	Suspicious	Completed At	Actions
System Administrator Raj	[REDACTED]	0% 0/80	✓ 0 / ✗ 20	1 	0	14 Apr 2026 03:39	View Details
chatgpt	[REDACTED]	95% 76/80	✓ 19 / ✗ 1	0	0	13 Apr 2026 10:36	View Details

Your Exam Results - 12 Apr 2026

Inbox

**ExamGuardian Proctor** 4 days ago

to me ▾

**ExamGuardian Proctor**

Your Exam Results Are Ready

Dear **Rajvardhan Singh**,Your exam has been successfully completed on **2026-04-12 14:40:47**.**5%****FAILED**

Roll Number: 20260412144047

Application ID: 123456789

Total Marks: 4 / 80

Correct Answers: 1

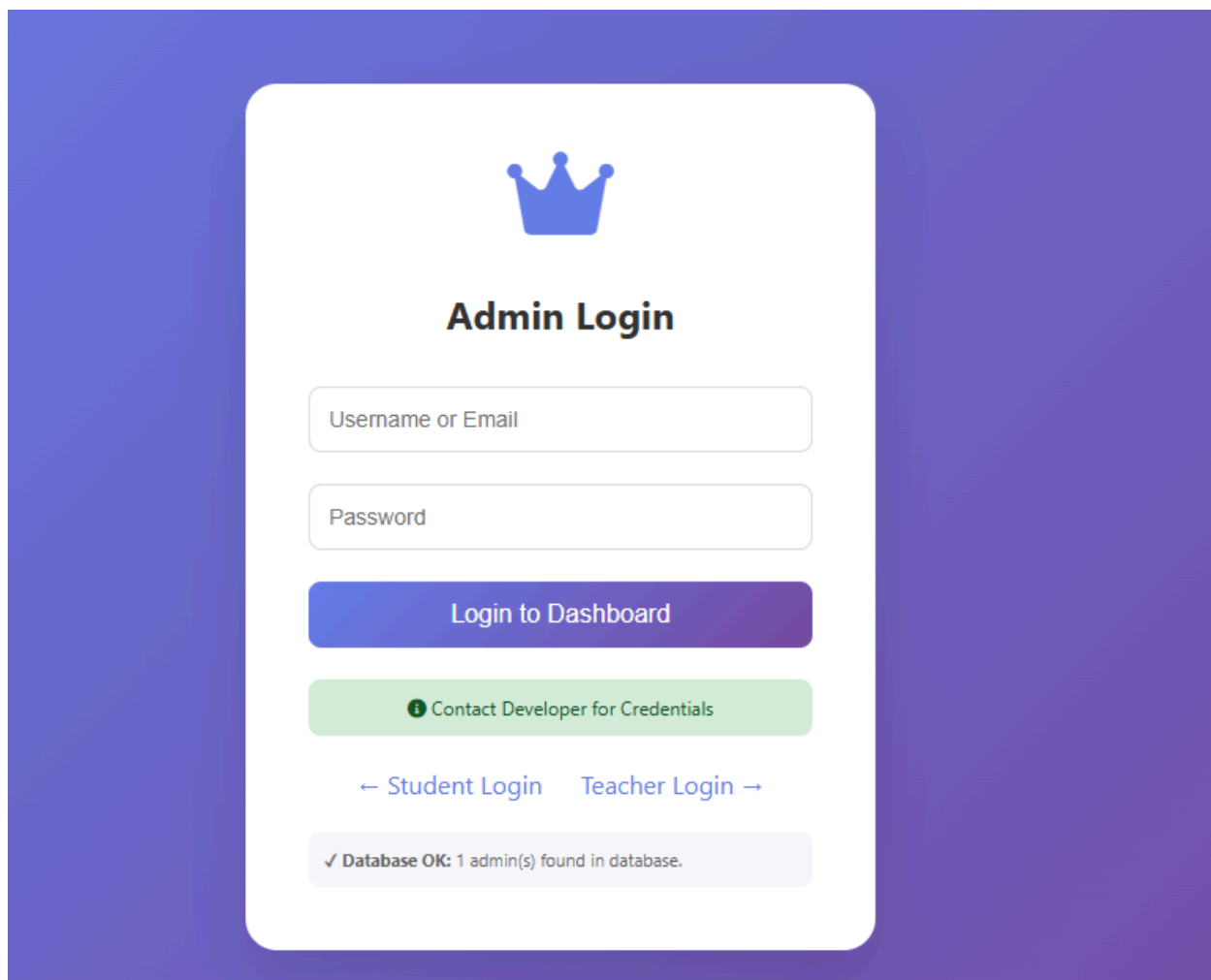
Wrong Answers: 19

Grade: F (Need Improvement)

Answer Summary (First 3 questions)

Administration & System Core

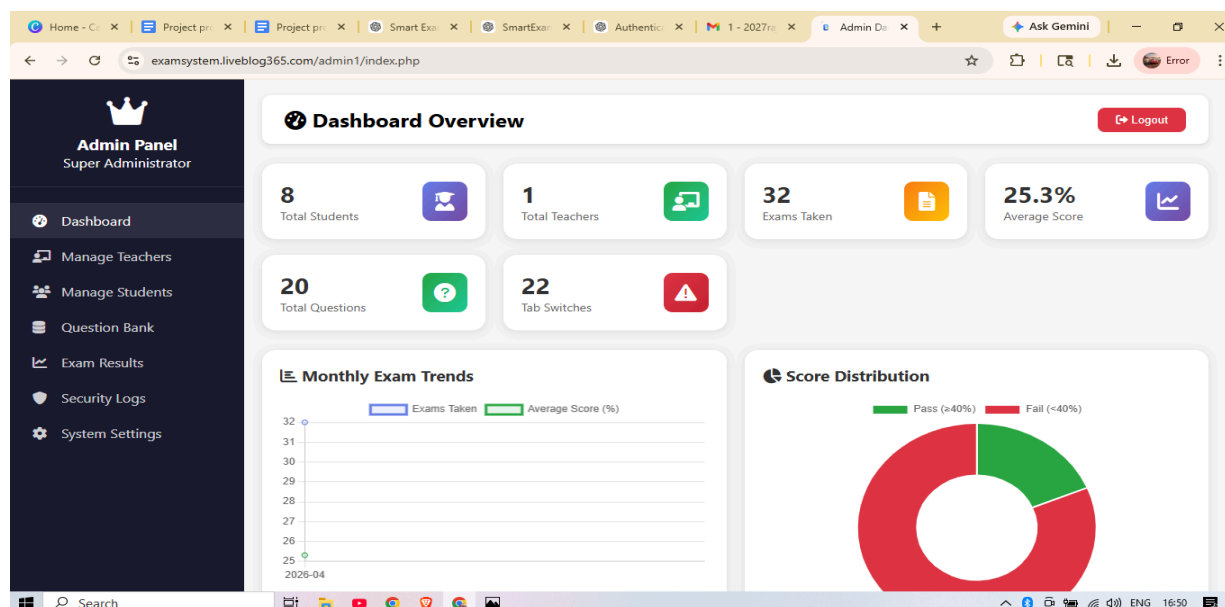
The **Administration Core** is the backbone of SmartExam, providing centralized control over system operations, user management, and security governance. It is designed to give administrators complete visibility and authority to ensure smooth, secure, and efficient functioning of the platform at scale.

The image shows a screenshot of the 'Admin Login' interface. It features a white login card centered on a purple gradient background. At the top of the card is a blue crown icon. Below the icon, the text 'Admin Login' is displayed in a bold, black font. There are two input fields: 'Username or Email' and 'Password'. Below these fields is a blue button labeled 'Login to Dashboard'. Underneath the button is a green button with an information icon and the text 'Contact Developer for Credentials'. At the bottom of the card, there are two links: '← Student Login' and 'Teacher Login →'. A status message at the very bottom of the card reads '✓ Database OK: 1 admin(s) found in database.'

Admin Dashboard

The **Admin Dashboard** offers a high-level overview of the entire system in real time. It displays key metrics such as the total number of students and teachers, along with system-wide analytics. This enables administrators to quickly assess platform usage,

monitor activity levels, and make informed operational decisions.



Exam Management

This module allows administrators to **plan and control the full exam lifecycle**.

- Schedule exams with specific dates and durations
- Assign exams to selected students or groups
- Configure timing, rules, and exam conditions (e.g., negative marking, attempt limits)

It ensures that exams are conducted in a structured and well-organized manner.

Security Controls

Security is a core priority within the Administration Core. This module provides:

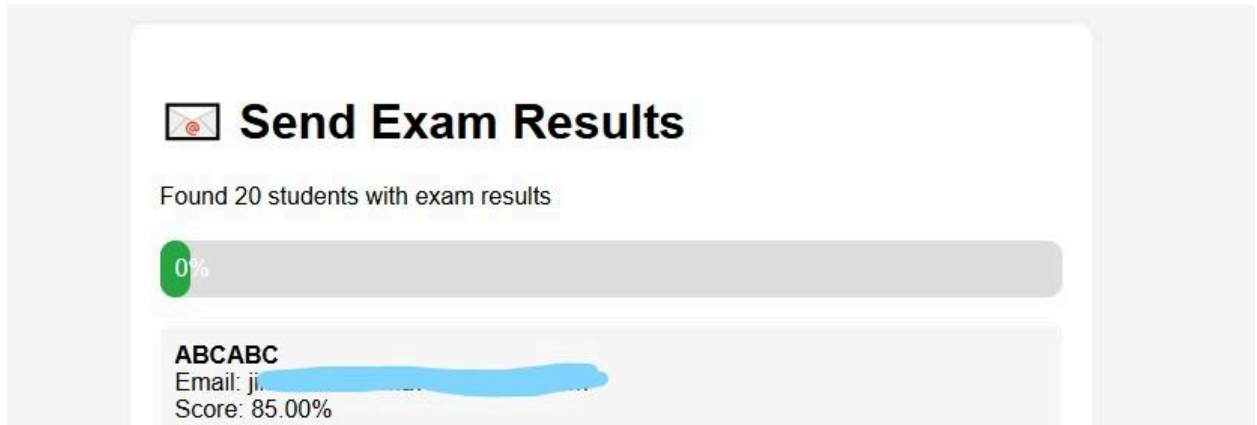
- **Audit Logs** to track all system activities and user actions
- **Cheating Detection Tuning** to adjust sensitivity of AI-based monitoring systems
- **Role-Based Access Control (RBAC)** to define permissions for admins, teachers, and students

These features ensure transparency, accountability, and strong protection against misuse.

Notifications

The notification system ensures smooth communication across all users:

- Create and manage **email templates** for different events (registration, exams, results)
- Configure **automated alerts** for important actions such as exam schedules, violations, and result announcements



This helps maintain timely and professional communication throughout the system.

Exam Results Analytics

The **Exam Results Analytics** module provides a **comprehensive performance overview** of all conducted exams. It enables administrators to track system-wide outcomes and make data-driven decisions.

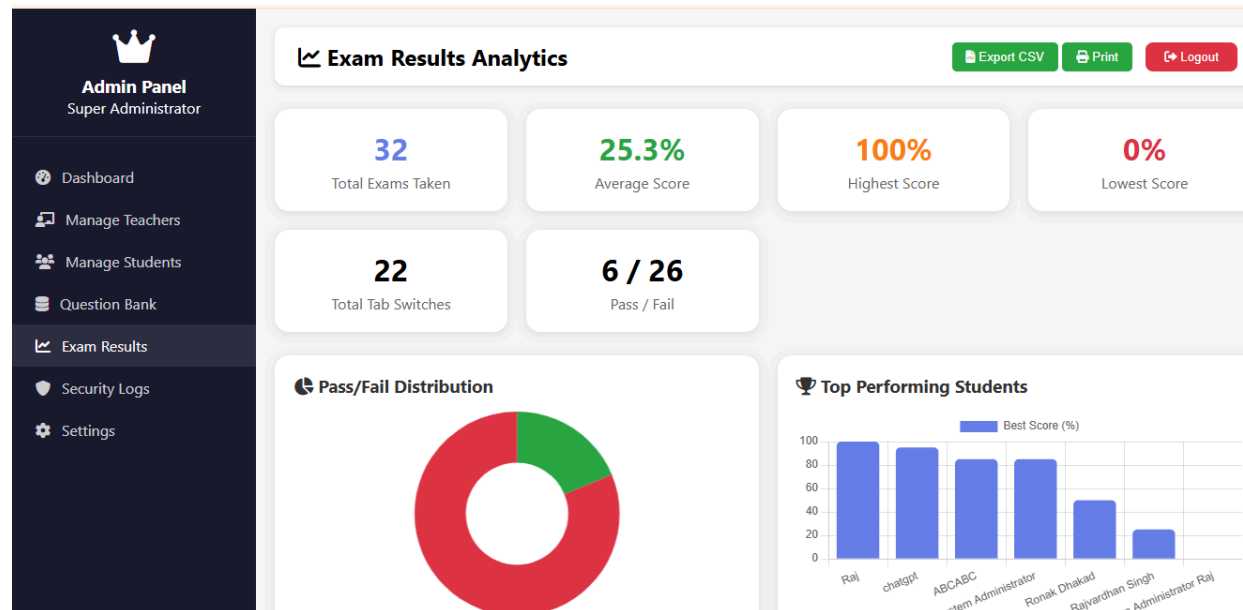
Key Highlights:

- Total Exams Taken (e.g., 32)
- Average Score (e.g., 25.3%)
- Highest & Lowest Scores (e.g., 100%, 0%)
- Total Tab Switches (e.g., 22)
- Pass/Fail Ratio (e.g., 6 / 26)

Advanced Features:

- **Pass/Fail Distribution Visualization** for quick insights
- **Top Performing Students Section** to identify high achievers
- **Search & Filters** (by name, roll number, date range, minimum score)
- **Detailed Result Table** including:
 - Student Name, Roll Number, Application ID
 - Score & Correct/Wrong Answers
 - Tab Switches & Suspicious Activity Flags
 - Completion Timestamp

- Quick Actions (View details)



Manage Students

This module allows administrators to efficiently **control and monitor student records** within the system.

Key Features:

- Overview metrics such as **Total Students** and **Exam Participation Count**
- Detailed student listing with:
 - Application ID & Roll Number
 - Full Name, Email, Phone Number
 - Registration Date

Actions Available:

- View complete student profile

- Delete or manage student records

Question Bank Management

The **Question Bank Management** module enables administrators to maintain a **centralized repository of exam questions**.

Key Metrics:

- Total Questions (e.g., 20)
- Total Marks (e.g., 80)
- Average Marks per Question (e.g., 4)

Functional Capabilities:

- Add new questions manually
- Bulk import questions for large-scale exams
- Edit or delete existing questions

Admin Power & Control

With these combined modules, administrators gain:

- Full visibility into system performance
- Complete control over users and exams
- Data-driven decision-making capabilities
- Efficient content and resource management

SmartExam's admin system is built to handle **real-world scale**, ensuring reliability, security, and operational excellence.

Intelligent Scoring & Evaluation System

Intelligent Scoring System

The **Intelligent Scoring System** in SmartExam is designed to deliver **accurate, transparent, and automated evaluation** of student performance. It ensures fairness while providing meaningful insights into exam outcomes.

Scoring Logic

Each question is assigned a **fixed weight of 4 marks**, ensuring consistency across all assessments.

To discourage guessing and improve answer quality, the system applies **negative marking of -1 for every incorrect response**. Unattempted questions do not affect the score.

Automated Evaluation

All responses are evaluated instantly by the system without manual intervention. The platform automatically calculates:

- Total score based on correct and गलत answers
- Final percentage
- Pass/Fail status (based on defined criteria)

This eliminates delays and ensures immediate result availability.

Performance Metrics

Beyond basic scoring, SmartExam provides **detailed performance insights**, including:

- **Accuracy Rate** (correct answers vs total attempted)
- **Score distribution** across exams
- **Behavioral indicators** such as tab switching impact

These metrics help both students and educators better understand performance patterns and areas for improvement.

Advanced Security & Monitoring System

Security & Integrity Measure

SmartExam is built with a strong focus on security, trust, and exam integrity, ensuring that every assessment is conducted in a controlled and fair environment. The system combines multiple layers of protection to safeguard user data and prevent malpractice.

Role-Based Authentication

Access to the platform is strictly controlled through role-based authentication, ensuring that students, teachers, and administrators can only access features relevant to their responsibilities. This minimizes unauthorized access and maintains system discipline.

Encrypted Credential System

All user credentials are securely stored using encryption mechanisms, protecting sensitive information such as passwords and personal data. This ensures data confidentiality and prevents unauthorized breaches.

Anti-Cheating Mechanisms

SmartExam integrates advanced anti-cheating features such as tab-switch detection, copy-paste restrictions, and AI-based monitoring. These measures actively discourage unfair practices and maintain the credibility of exams.

Real-Time Monitoring

The system continuously monitors user activity during exams, tracking

behavior and detecting anomalies in real time. Any suspicious actions are flagged instantly, allowing for quick review and intervention.

Audit Trail Logging

Every critical action within the system is recorded through detailed audit logs. This provides complete transparency, enabling administrators to trace activities, investigate issues, and ensure accountability across all users.

Future Enhancements

SmartExam is designed with a forward-looking approach, allowing continuous upgrades to meet evolving educational and security needs. The following enhancements aim to strengthen intelligence, accessibility, and trust within the platform.

AI Face Recognition Improvement

Future updates will enhance the accuracy and reliability of face recognition systems. This includes better identity verification, improved liveness detection, and the ability to handle varied lighting conditions and multiple environments, ensuring more precise and secure proctoring.

Voice Monitoring System

A voice-based monitoring layer will be introduced to detect unusual audio activity during exams. This feature will help identify conversations, external assistance, or suspicious background noise, adding an extra level of exam integrity beyond visual monitoring.

Mobile App Integration

SmartExam plans to expand into mobile platforms, enabling students and educators to access exams, dashboards, and notifications through dedicated mobile applications. This will improve accessibility, flexibility, and user engagement across devices.

Blockchain-Based Result Verification

To enhance trust and transparency, blockchain technology can be integrated for **tamper-proof result storage**. This will allow institutions and employers to verify certificates and results securely, eliminating the risk of data manipulation or fraud.

Advanced Analytics Dashboard

Future versions will include a more powerful analytics engine with interactive dashboards. These will provide deeper insights such as predictive performance analysis, learning trends, and AI-driven recommendations for both students and educators.

Conclusion

ExamGuardian Proctor represents a complete, AI-driven examination ecosystem built to address the core challenges of modern online assessments. By combining intelligent proctoring, automated evaluation, and real-time monitoring, the platform ensures a high level of reliability and trust in digital exams.

The system is designed to deliver:

- **Transparency** through detailed analytics and audit logs
- **Security** via advanced anti-cheating and authentication mechanisms
- **Automation** with instant evaluation and result processing
- **Scalability** to support institutions of varying sizes and demands

With its robust architecture and future-ready capabilities, ExamGuardian Proctor is an ideal solution for **colleges, universities, and competitive examination platforms**, offering a secure, efficient, and modern approach to conducting online examinations.

Acknowledgment & Thank You

I would like to express my sincere gratitude to everyone who contributed to the successful development of this project, **SmartExam / ExamGuardian Proctor**. This project is not just a technical implementation but a result of teamwork, learning, and continuous effort.

First and foremost, I would like to thank our **faculty members and mentors** for their guidance, valuable suggestions, and constant support throughout the development process. Their insights helped shape the project into a more structured and practical solution.

I would also like to appreciate all the **team members** from both the Cyber Security and Computer Science domains. Each member played an important role, whether in

development, testing, security implementation, or documentation. The collaboration, dedication, and problem-solving approach of the team made this project possible.

A special thanks to the **open-source community and technologies** such as OpenCV, TensorFlow, and modern web development tools, which provided the foundation for building advanced features like AI proctoring and real-time monitoring.

I am also grateful to everyone who **tested, reviewed, and provided feedback** on the system. Their suggestions helped us improve usability, performance, and reliability.

Finally, I would like to thank everyone who supported this project directly or indirectly. This project reflects our collective effort, learning, and passion for building innovative solutions in the field of online education.

Thank You 🙏

Rajvardhan Singh Badgujar